GPS-Based Customer Behaviour Tracking & Staff Assistance

1. Component Overview

The GPS-Based Customer Behaviour Tracking and Staff Assistance system is an intelligent indoor monitoring platform designed to autonomously track customer movements within a retail environment. By capturing high-frequency GPS signals from a customer's mobile device, the system leverages Machine Learning to classify the customer's behavioral intent—distinguishing between active browsing and mere transiting. When the system detects behavioural patterns indicative of confusion, prolonged searching, or pacing, it proactively dispatches automated alerts to store personnel, ensuring timely assistance without requiring the customer to actively request help.

2. Problem Addressed

In large and busy retail environments, store staff struggle to identify which customers require assistance. Traditional approaches rely on manual observation, which is inherently subjective, inconsistent, and unscalable. Consequently, customers who are unable to locate specific products or sizes often become frustrated and abandon their shopping journey, leading to lost sales and diminished customer satisfaction. Furthermore, many customers hesitate to actively seek out staff for help, meaning their needs remain completely unaddressed.

3. Main Objective

The main objective of this component is to develop an automated, intelligent tracking and alerting system that passively monitors customer movement inside a store, classifies their shopping behaviour using machine learning, and proactively alerts staff when a customer demonstrates a need for assistance.

4. Specific Objectives

- To implement a reliable indoor spatial tracking mechanism using mobile GPS hardware, compensating for signal inaccuracies with 3D boundary tolerance.
- To train and deploy a Machine Learning model capable of classifying real-time customer movement into distinct behavioural intents ("Browsing" vs. "Transiting").
- To design a historical path-tracking algorithm that identifies complex behavioural anomalies, such as repetitive pacing between zones.
- To develop an automated alerting engine that triggers staff notifications based on prolonged dwell times and pacing patterns.
- To construct a real-time operational dashboard for store staff to monitor live customer locations and manage assistance requests.

5. Main System Functions

- Real-Time Location Tracking: Continuous background extraction of GPS coordinates (latitude, longitude, and altitude) from the customer's mobile device at 3-second intervals.
- Zone Boundary Resolution: Dynamic matching of raw GPS coordinates to predefined 3D virtual store sections (zones).
- Intent Classification: Real-time Machine Learning prediction of customer behaviour based on movement features (velocity, direction change, dwell time).
- Automated Alert Generation: Autonomous generation of staff assistance requests triggered by AI-detected "Browsing" dwell time ($\geq$ 30 seconds) or algorithmic pacing detection (Zone A $\rightarrow$ Zone B $\rightarrow$ Zone A within 60 seconds).
- Staff Dashboard Management: A live interface for staff to view customer locations and resolve active assistance requests.

6. System Workflow

The operational workflow follows a continuous, real-time loop:

1. Initialization: The customer opens the mobile app, which immediately begins transmitting a GPS heartbeat every 3 seconds to the backend server.

2. Zone Mapping: The backend resolves the coordinates against predefined 3D store zones (applying a ±100m altitude tolerance).

3. Feature Extraction & Prediction: The server calculates distance (via the Haversine formula), velocity, and time elapsed, passing these features to a Random Forest model to predict intent.

4. State & History Update: The system logs the customer's current zone and appends any zone transitions to a chronological history array.

5. Evaluation: The alerting engine evaluates the current state against dwell-time thresholds and the history array against pacing pattern rules.

6. Notification: If a threshold or pattern is breached, an assistance alert is recorded in the database, which is immediately reflected on the staff dashboard via high-frequency polling.

7. System Architecture

The system adopts a highly decoupled, three-tier architecture:

• Data Acquisition (Client Tier): A mobile application utilizing the device's native geolocation hardware to capture coordinates.

• Processing & Orchestration (Middle Tier): A backend that handles point-in-zone geometry calculations, Haversine distance computations, and hosts the serialised Machine Learning models for rapid inference.

• Persistence (Data Tier): A NoSQL database optimized for real-time document updates, storing zone boundaries, live tracking sessions, historical paths, and staff alerts.

8. Technologies Used

- Frontend / Mobile App: Cross-platform mobile framework (Flutter), Geolocation packages
- Backend API: Python, FastAPI REST framework
- Machine Learning: Scikit-Learn, Pandas Dataframes
- Database: Firebase Firestore (NoSQL Document Store)
- Mathematical Algorithms: Haversine Formula (Great-circle distance), 3D Axis-Aligned Bounding Box (AABB) collision detection

9. Dataset and Data Scope

The Machine Learning model was trained on a synthetic customer movement dataset comprising records of simulated retail navigation. The data encompasses varying walking speeds, angular direction changes, and zone dwell durations.

• Scope: The dataset is strictly limited to spatio-temporal movement features and does not contain personally identifiable information (PII).

• Preprocessing: The dataset was split into 80% training data and 20% testing data using a fixed random state for reproducibility.

10. Backend Implementation

The backend is structured into modular services:

• Monitoring Routing: The core orchestration router that processes incoming GPS heartbeats, manages the state machine, and orchestrates calls to the alerting engine.

• Zone Spatial Service: Implements spatial logic. It constructs minimum/maximum boundary boxes from four recorded corner points and executes point-in-zone containment checks. Due to indoor altitude unreliability, a static ±100-metre vertical buffer is applied during geometry resolution.

• Trajectory Machine Learning Service: Manages feature engineering (Haversine distance calculation) and executes the Random Forest model for inference. It includes a rule-based fallback mechanism (speed > 0.5 m/s implies transiting) to guarantee system resilience if

the model fails to load.

11. Frontend Implementation

The frontend is divided into two distinct user roles:

• Customer Application: Utilizes periodic background timers and geolocation packages to silently poll and transmit GPS coordinates every 3 seconds. It also features a manual SOS button for overriding automated detection.

• Staff Dashboard: Implements a real-time polling mechanism (3-second intervals) to fetch active tracking sessions and pending alerts, rendering a live operational view of the retail floor.

12. Database Implementation

The NoSQL database is structured around four primary collections:

1. Zones: Stores physical section boundaries (min/max coordinates).

2. Customer Monitoring: Maintains the live state of each customer, including their current velocity, predicted intent, and a historical array containing timestamped zone transitions.

3. Assistance Requests: Records pending and resolved help alerts, including metadata on the trigger type (pacing, dwell, or manual).

4. Staff Notifications: A decoupled notification log for individual staff devices.

13. Implementation Results

The implementation resulted in a fully functional, end-to-end prototype capable of accurately resolving indoor locations to predefined virtual zones using standard mobile hardware. The integration of the Machine Learning pipeline successfully enabled the system to transition from basic spatial tracking to complex behavioral understanding. The alerting engine correctly fired notifications in real-time, validating the architectural design and demonstrating the feasibility of autonomous retail assistance.

14. Machine-Learning Results

The chosen algorithm for behavioural classification was a Random Forest Classifier comprising 100 decision trees. Ensemble methods like Random Forest proved highly robust against the inherent noise and fluctuations present in indoor GPS data. The model successfully distinguished between the erratic, slow-moving patterns of "Browsing" and the direct, high-speed patterns of "Transiting."

15. Important Features Identified by the Models

- Velocity (m/s): The primary differentiator; transiting customers exhibited consistently higher velocities than browsing customers.
- Distance Moved (meters): Extracted via the Haversine formula, providing a standardized metric for physical displacement between consecutive heartbeats.
- Direction Change (degrees): Browsing behaviour is characterized by frequent, sharp angular changes, whereas transiting usually follows a linear trajectory.
- Zone Dwell Time (seconds): Extended duration within a single zone strongly correlated with browsing intent.

16. Functional Testing Results

- Spatial Resolution: The point-in-zone detection performed reliably, with the ±100m altitude tolerance successfully mitigating false negatives caused by vertical GPS drift.
- Intent Prediction: The ML model rapidly classified behaviour changes (e.g., stopping to look at a shelf) within one to two heartbeat cycles.
- Alert Triggering: The dwell-time logic accurately fired after 30 seconds of continuous browsing, and the complex array logic successfully identified A $\rightarrow$ B $\rightarrow$ A pacing patterns in real-time.

17. Research Contribution

This research contributes a novel methodology for indoor customer behaviour tracking by demonstrating that standard, unaugmented mobile GPS hardware—when combined with intelligent geometric tolerance (altitude buffering) and Machine Learning—can achieve sufficient accuracy to drive an automated behavioural alerting engine. Unlike traditional approaches requiring expensive indoor beacon infrastructure (Bluetooth/RFID) or privacy-invasive camera systems, this system operates entirely on ubiquitous mobile hardware and passive spatial algorithms.

18. Current Limitations

- Indoor GPS Accuracy: Standard GPS signals degrade significantly indoors. While altitude tolerance mitigates vertical drift, horizontal drift can still cause edge-case misclassifications when customers stand near the physical border of two zones.
- Battery Consumption: Continuous, high-frequency (3-second) background GPS polling significantly impacts the mobile device's battery life.
- Synthetic Dataset Reliance: The Machine Learning model was trained on synthetic data. While it performs well theoretically, real-world customer movement is highly chaotic and may require model retraining on empirical data to maintain accuracy in a live production environment.

- Dependency on Active App: The system requires the customer to keep the store application open and running in the background to transmit location heartbeats.